Reg. No. | | | | | | | | | | |

[14CS/IS33]

THIRD SEMESTER B.TECH EXAMINATION DECEMBER–2015

COMPUTER ORGANIZATION

Time:3 Hours **Maximum Marks:70**

Instruction: Answer any **FIVE** full Questions, choosing at least **TWO** from each Part.

PART – A

1. a. Explain the basic operation of the processor with neat diagram. **07**

   b. What are factors which are used to judge the performance of a computer? Explain any four of them. **07**

2. a. Register R_1 and R_2 of a computer contain the decimal values 1800 and 3300. What is the effective address of the memory operand in each of the following instructions?

   i. Load $60(R_1),R_4$ ii. Mov #2000, R_5

   iii. sub $(R_1)+$, R_5 iv. Add--(R_2), R_5 **08**

   b. Write a assembly language program to find the sum of number from 1 to 100 using indirect addressing mode. **06**

3. a. Explain basic I/O operation involving a keyboard and display a device. Write a assembly language program to read one line from the keyboard, store in the memory buffer and echoes it back to the display. **08**

   b. Explain the logical instructions with an example to each. **06**

4. a. Define vector Interrupt. **03**

   b. Explain the different ways of disabling interrupt. **04**

   c. Which types of I/O devices are interfaced through DMA? Explain the bus-arbitration process used for DMA. **07**

PART – B

5. a. How cache blocks are mapped to main memory module using direct mapping method and associate mapping method. **06**

   b. How many chips are needed to provide a memory capacity of 64KB? Explain with neat diagrams how the chips are to be connected to the address bus? Specify the side of decoders. **06**

   c. What do you understand by Flash memory? **02**

6. a. Multiply 10111 and 01001 using Booths algorithm. **06**

   b. What are the advantages of Cary-look ahead adder? Explain CLA adder with neat diagram and boolean equation. **08**

7. a. If A and B are two single precision floating point numbers where:

   A=4ABCDEF7H

   B=5EDCBA96H

   Show the result of (A+B) and (A*B). **08**

   b. Explain the restoring division algorithm with an example. **06**

8. a. Write a short note on microwave oven. **06**

   b. Write the control sequence for the instruction sub(R_2), R_1. **08**

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